

EXP 6 Write a Servlet to demonstrate session tracking using HttpSession.

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- ☐ The user will enter their username and password in a login form.
- ☐ The servlet will validate the credentials and create an HTTP session for the user if authentication is successful.

2. Steps for Implementing Session Tracking with HttpSession:

1. Create the Login Form (HTML):

- o Create an HTML form with two input fields for username and password.
- o Provide a submit button to send the data to the servlet for authentication.

2. Create the Login Servlet:

- o Write a Java servlet class to handle login requests.
- o The servlet will:
 - ☐ Retrieve the username and password from the form.
 - ☐ Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).
 - ☐ Create a new HTTP session for the user if the credentials are valid.
 - ☐ Set session attributes such as the user's username to track the session.

3. Validate User Credentials:

- o In the servlet, check if the username and password match the expected values.

- o If valid, create a session and store the username in the session.

- o If invalid, redirect the user back to the login page with an error message.

4. Create and Track User Session:

- o Upon successful login, use the HttpSession object to store the session data.

- o Use the session.setAttribute("username", username) method to save the username.

- o Optionally, set session timeout using

session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.

5. Redirect to a Protected Page:

- o After a successful login, the user will be redirected to a protected page (e.g., a dashboard or home page).

- o The protected page should check if the user is authenticated by checking the session attribute.

6. Logout and Session Invalidation:

- o Provide a logout feature that invalidates the session using session.invalidate() when the user logs out.

- o After logout, redirect the user back to the login page.

7. Session Expiration:

- o If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

OUTPUT:

Session Tracking Demo: Login System

Login Form

Username:

Password:

Login

Active Session Status

User Logged In:	Yes (Test User)
Session ID:	1234abcd-56ef-78gh-91ij
Session Creation Time:	2022-10-27 17:54:39.097
Last Accessed Time:	2022-10-27 17:54:39.317
Visit Count:	3
Data Stored in Session:	{key1: value1, username: testuser}
Logout	

This Servlet uses `HttpSession` to manage user logins. The unique Session ID is tracked to maintain user state. You can view the data stored and the visit count (a data point updated by the Servlet) in the table. Logging out will invalidate the session and remove all data.

RESULT:

A simple login system is created where the user's session is tracked by a servlet to demonstrate session tracking using `HttpSession`.